

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 22

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Interconnected biological processes recycle nutrient elements so small amounts support life indefinitely without depletion; waste products of one process are raw materials of another so wastes do not build up (1) • Transpiration returned water vapour to the atmosphere increasing rainfall elsewhere (1) • Allowing more plant growth, further stabilising climate and soils (1)	3
01.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Lack of ancient historical / instrumental data (1) • Reliance on proxy data (tree rings, pollen) of limited reliability for ancient conditions (1) • Lack of sophisticated equipment / inability to measure many factors (1) • Limited coordination between researchers; lack of data collection in many areas (oceans, poles) (1)	3
01.3	<i>1 mark for each point.</i> • The ratio of oxygen-18 to oxygen-16 in the ice indicates the temperature when the snow fell (1) • Air bubbles trapped in ice give the composition of the past atmosphere (CO₂, methane) (1) • Ratio of carbon-14 to carbon-12 dates organic material (eg pollen in sediments) (1) • More magnesium is incorporated into calcite at higher temperatures, so the ratio estimates past sea temperature (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 8-9; AQA spec 3.1.1.2-3.1.1.3.	

Question	Marking guidance	Marks
02.1	<i>1 mark for each point.</i> - The new opportunity is overwhelmingly attractive (economic benefit) (1) - Lack of knowledge / understanding of the impacts (1) - The impacts occur elsewhere or much later, so are not evident to those who benefit (1)	3
02.2	<i>1 mark for each point.</i> - Cultural belief that resources should be exploited to satisfy needs only (1) - Problems of over-exploitation become obvious quickly as local supplies dwindle, so behaviour changes (1)	2
02.3	<i>1 mark for each point.</i> - Resources are brought in from other areas (1) - If one area is depleted, exploitation moves to another area (eg timber, fish, metals, energy) without the consumers noticing (1)	2
02.4	<i>1 mark for each point.</i> - Easier access to energy has made many new activities possible (1) - The scale of activities has greatly increased (1) - Toxic wastes are produced that cannot be decontaminated by natural processes (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-371; AQA spec 3.6.1.	
03.1	<i>1 mark for each point.</i> - Silver carp eat phytoplankton, bighead carp zooplankton, grass carp and tilapia vegetation, common carp bottom-feed on molluscs / insects, catfish or perch eat small fish (1) - Species occupy different niches / trophic levels so they do not compete and all food sources are used (1) - Bottom feeders re-suspend nutrients boosting plankton / plant growth (1) - Predators control over-breeding species (tilapia) so remaining fish grow larger; total yield exceeds any single species (1)	4
03.2	<i>1 mark for each point.</i> - IMTA: fed species (salmon, shrimp) alongside inorganic extractive species (seaweeds absorbing ammonia and phosphate) and organic extractive species (filter-feeding mussels eating waste particles / plankton) (1) - Wastes of one species become nutrients for another so less pollution and extra harvests (1) - Aquaponics: aquaculture water rich in nutrients / organic matter feeds hydroponic salad crops (lettuce, herbs, watercress) (1) - Plants remove the nutrients that would cause deoxygenation and cleaned water returns to the fish (tilapia, carp, catfish); rice-fish paddies similar (1)	4
03.3	<i>1 mark for each point.</i> - Bluefin tuna do not breed well in captivity so young fish are caught from the wild and fattened in sea cages (1) - Production still depends on (and removes juveniles from) the wild population; carnivores need large fish inputs (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 357-358; AQA spec 3.5.2.3.	

Question	Marking guidance	Marks
04.1	<i>1 mark for each point.</i> • k-selected: very slow growth, does not breed until 30, few eggs, so the population cannot replace losses (1) • Fish reach catchable size (20) before they breed (30) so fishing can remove the entire future breeding population (1) • Deep-water aggregations easily located and caught by trawling (1) • Recovery takes decades so fisheries have had to be closed (1)	4
04.2	<i>1 mark for each point.</i> • Herring breed at 2 years - the same age they become catchable - so there is always a breeding population and many young (1) • Cod are catchable at 2 but do not mature until 4, so heavy fishing removes most fish before they breed (1)	2
04.3	<i>1 mark for each point.</i> • Larger numbers of smaller / younger fish are caught, or bigger boats, nets and sonar increase catch effort (1) • Masks the decline of the breeding population until a sudden collapse (1)	2
04.4	<i>1 mark for each point.</i> • Neither is a target species but both are caught in demersal trawls; skate now extinct in the Baltic and critically endangered elsewhere (1) • Very late maturity (skate 11 years, Greenland shark 100-150 years, living over 400) and few young mean even low by-catch mortality exceeds replacement (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 341-342, 348-349; AQA spec 3.5.2.2.	
05.1	<i>1 mark for each point.</i> • The range of an environmental factor within which a species / population can survive (survival highest at the optimum, falling towards the limits) (1) • X survives from about pH 5.5 to 8.5 (wide range); Y only from about pH 6.0 to 8.0 (narrow range) (1) • Both have the same optimum (pH 7.0) (1)	3
05.2	<i>1 mark for each point.</i> • X: 60% survive at pH 6.0 whereas only 5% of Y survive (1) • X has a larger gene pool / wider range of tolerance so some individuals are adapted to the new conditions (1) • Survivors breed and their offspring are adapted so the population rebuilds; Y is likely to die out locally (1)	3
05.3	<i>1 mark for each point.</i> • Reveals the habitat requirements / which abiotic factor is near the edge of tolerance (1) • Management can make small beneficial changes to that factor (eg liming, shading, water levels) to keep conditions within the range and increase survival (1)	2
05.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Food supply (1) • Pollinators (1) • Seed dispersal species (1) • Disease / pathogens (1) • Predators / competitors (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 72-73; AQA spec 3.1.3.1.	

Question	Marking guidance	Marks
06.1	<i>Correct answer only.</i> • The readings are precise but may not be accurate (1)	1
06.2	<i>1 mark for each point.</i> • A result that differs from other results with which it was expected to be similar (1) • Repeat the study / measurement: if it recurs it is a 'real' result caused by another variable; if not it was an error and can be excluded (1)	2
06.3	<i>1 mark for each point.</i> • To test the reliability of the method / equipment (1) • To decide the sample size, number of samples, spacing or timing needed to detect the variation without collecting unnecessary data (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-391; AQA spec 3.7.1.	
07.1	<i>1 mark for each point.</i> • Timber: building construction, furniture, tools, telegraph poles, concrete shuttering (oak, beech, mahogany, teak, conifer softwood) (1) • Fuel: fuelwood and charcoal - still the main energy source for most people in LEDCs (1) • Food: fruit, nuts, bushmeat, livestock fodder; crops of forest origin (coffee, cacao, bananas, Brazil nuts); pigs and chickens were forest animals (1) • Fibres: paper (wood pulp cellulose), cotton, viscose / rayon (1) • Medicines: quinine (cinchona), taxol (yew), aspirin (willow); most species not yet researched (1)	5
07.2	<i>1 mark for each point.</i> • Wood is mainly cellulose which is hard to digest, so carbon stored in wood has a long residence time; total forest carbon (vegetation plus soil) is about double that in the atmosphere (1) • Growing trees sequester CO ₂ by photosynthesis and release oxygen; forest soils hold more carbon than the trees (especially boreal) (1) • Planting / protecting forests removes CO ₂ and reduces the enhanced greenhouse effect (1)	3
07.3	<i>1 mark for each point.</i> • Highest transpiration of any ecosystem increases downwind rainfall; interception increases evaporation; deep organic soils retain water and delay runoff so river flow is more even and flooding reduced (1) • Low albedo absorbs sunlight; heat stored and released at night reduces day-night temperature extremes; canopy microclimate: lower light, wind, higher humidity (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-364; AQA spec 3.5.3.1.	
08.1	<i>1 mark for each point.</i> • Images / maps of large areas; land-use change, deforestation, flooding (1) • Vegetation density and surface temperature (1) • Wind velocity, sea-surface altitude, wave height, oil pollution (1)	3
08.2	<i>1 mark.</i> • Microwaves pass through clouds (unlike visible light) (1)	1

Question	Marking guidance	Marks
08.3	<i>1 mark for each point.</i> • Very large / remote areas can be surveyed quickly and repeatedly so changes over time are detected (1) • No need for researchers to visit inaccessible areas, reducing cost and disturbance; data are standardised across the whole area (1)	2
08.4	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Motion-sensitive cameras placed along trails record animals passing, using visible light by day and infrared at night (1) • Each tiger has a unique pattern of facial / body stripes (1) • Images are compared with the database so individuals are recognised rather than counted twice (1) • The number of different individuals recorded gives a minimum population; repeated captures give territory size, movements, lifespans and social groupings (1) • Limitation: only works for species with unique visible features (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-403, 406-407; AQA spec 3.1.2.3.6.	
09.1	<i>1 mark for each point.</i> • Upwelling currents bring nutrients from the seabed to surface waters (1) • 24-hour daylight in summer allows a huge phytoplankton bloom (1) • Algae also grow under sea ice where krill shelter and feed; krill support most of the food web (1)	3
09.2	<i>1 mark for each point.</i> • Krill is the main food of baleen whales, penguins, seals and fish so over-harvesting removes food from most of the web (1) • Toothfish overfishing removes the food of sperm whales and elephant seals (1) • Long-lining for toothfish catches and drowns albatrosses on the hooks (by-catch) (1)	3
09.3	<i>1 mark for each point.</i> • Ice on land stores 70% of the Earth's fresh water, keeping sea levels low (1) • High albedo of ice reflects sunlight, limiting warming (reduction in ice area causes further warming) (1) • Carbon sequestration: carbon in algae eaten by krill sinks to the seabed in krill faeces (1)	3
09.4	<i>1 mark.</i> • Stronger winds spreading ice; more snowfall from warmer, moister air freezing onto slush; fresh meltwater from land forming a surface layer that freezes more easily (long-term trend still expected to be a decline) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 67-71; AQA spec 3.1.2.3.4.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • In-situ: species remain in natural conditions with the whole community and inter-species relationships; protected areas (SSSI, NNR, SAC, SPA, Ramsar, MCZ), land ownership, agri-environment schemes, habitat creation and design (area, shape, corridors, diversity), management of abiotic and biotic factors, plagioclimax maintenance, population control, reintroductions and rewilding (Yellowstone wolves, Scottish beavers) • Limits of in-situ: external threats (pollution, climate change), enforcement, cost, conflicts with land use, small fragmented populations • Ex-situ: captive breeding when in-situ is impossible or insufficient; selection criteria; difficulties keeping and breeding (habitat size, food, relationships, breeding triggers, group dynamics, gene pool / inbreeding, hybridisation); techniques (stud books, cryopreservation, AI, embryo transfer, micropropagation, cloning); seed banks; successes (Hawaiian goose, Iberian lynx, red kite, large blue) • Limits of ex-situ: expensive, few species, loss of wild behaviour, small gene pools, release requires suitable habitat and post-release support (hard vs soft release, monitoring) • Judgement: ex-situ is a temporary safeguard / support that only succeeds if in-situ habitat exists to return to; both needed, in-situ fundamental	9
10.2	1 mark. • Restoring natural processes / communities to a habitat changed by humans, often by reintroducing missing (keystone) species and removing human management such as grazing (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-49; AQA spec 3.1.2.3.3-3.1.2.3.4.	

Question	Marking guidance	Marks
11.1	Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Ecological features: cnidarian polyps, cilia, nematocysts, zooxanthellae symbiosis, limestone skeletons building reefs, complex structure and food web; specific abiotic requirements (temperature, light, salinity, low turbidity) • Importance: quarter of marine species, fisheries, medicines, carbon sink, erosion protection, tourism • Vulnerability: narrow abiotic tolerance means sensitivity to warming, acidification, sedimentation, nutrient pollution, oil, bleaching; slow growth so slow recovery; physical damage; overfishing and food-web effects (crown of thorns); introduced species; dependence on mangroves and seagrass • Conservation: MPAs, No Take Zones, fishing controls, tourism management, protection of linked habitats, CITES • Comparison with other habitats (rainforests, mangroves, Antarctica) where relevant • Evaluation of local vs global threats and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4.	25

Question	Marking guidance	Marks
11.2	Discuss the extent to which agriculture can be made sustainable while feeding a growing human population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Need: population possibly 12 billion; humans raise carrying capacity; trophic level of diet • Impacts to be reduced: habitat loss, drainage, nutrient enrichment, introduced species, pesticide and nutrient pollution, nitrate toxicity, greenhouse gases, hydrological changes, soil erosion • Social, economic and political influences: consumer choices, technology availability, grants, guaranteed markets, surpluses, quotas, set-aside, agri-environment schemes • Strategies: cultural and biological pest control, integrated control, organic nutrient supply, legumes, low tillage, water conservation, CWR conservation, renewable energy, methane reduction, habitat retention • Limits: little new land, imports reducing local availability, cost and acceptance of new techniques, past land degradation • Justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-337; AQA spec 3.5.1.3-3.5.1.5.	25